There's a new boss in town
Leaked benchmarks reveal NVIDIA RTX 3060Ti outperforms the RTX 2080 Super.
https://dgit.in/dec20-39

ASRock Mars 4000U
ASRock has launched the thinnest AMD-powered mini-PC with support for up to a Ryzen 7 4800U. **https://dgit.in/dec20-40**

AMD Ryzen 9 5950X

Refined silicon incoming

AMD processors have been making huge improvements every year, be it in the form of increasing core counts or improving power efficiency. While the Ryzen processors have made massive IPC improvements over the older Bulldozer and Piledriver microarchitectures in the initial years, Intel still had the leadership in that domain. With Zen 3, a.k.a. Vermeer, a.k.a. Ryzen 5000 series, AMD is finally taking leadership in the last remaining bastion of performance metrics. Team Red is claiming a 19 per cent IPC improvement with Zen 3 over Zen 2 which is quite a notable achievement considering that AMD has not switched to a smaller node but has managed to include improvements to power efficiency and performance which are otherwise normally obtained with a node change.

Price
₹73,999

83

PERFORMANCE..........90
VALUE FOR MONEY....75

What's new

AMD has overhauled several aspects of the Zen 2 microarchitecture to arrive at Zen 3. For starters, there's the way the cores are arranged and linked within the die. With Zen 2, each compute die (CCD) had two core complexes (CCX) and each CCX had four cores / eight threads which were given a 16 MB L3 cache to work with. In Zen 3, each CCD has a single unified CCX with eight cores / 16 threads and 32 MB of L3 cache. Also, each package is limited to 2 CCDs so the maximum core count that you'll see with Zen 3 is 16 which is the same as Zen 2 which means we can't expect higher core counts this generation.

Performance

The AMD Ryzen 9 5950X has to do very little to be deemed better than the competition so the following results should hardly be a surprise. There's still scope for improvement with the core clock speeds and we might see the "XT" processors if AMD wants to bring them out. Also, Intel's Rocket Lake is around the corner so that's an additional reason for us to expect more from Zen 3 in the months to come.

As always, we begin with Cinebench 20 to see how the processor fares in single threaded and multithreaded performance. In the single-threaded benchmark, The AMD Ryzen 9 5900XT ends up with a massive lead over its own Zen 2 processors as well as Intel Core processors. It scores are in the 630-640 range which is the first time that we've ever seen anything go this high. The multithreaded run obviously scales really well and we can see the same in the results.

Our Handbrake run basically consists of transcoding a 4K video clip from one CODEC to another without changing the resolution. Handbrake is quick to incorporate new CPU encoding engines and is favoured among the 'creator community'. The 5950X shows a 17 per cent reduction in encode times compared to the Ryzen 9 3900X and about 10 per cent compared to the 10900K.

Blender is a fairly well known open-source rendering software. In this benchmark, we're looking at the time taken for a run to finish rendering (lower scores are better). The AMD Ryzen 9 5950X scores a significant lead over the 3900X as well as the 10900K – 286 seconds over 454 for the Intel 10900K. However, this difference is only brought out when running multiple iterations of the same workload. In a single run, the difference is barely there.

The 7-zip decompression benchmark places the Ryzen 9 5950X at the top with 123,436 MIPS (million instructions per second). It scales quite well and we see a significant lead over the 3900XT (103,240 MIPS) as well as the 10900K (84,182 MIPS).

In our gaming benchmarks, the AMD Ryzen 9 5950X performs at par or better in most games. Essentially, we're looking at the best performing gaming processors right here.

Verdict

This is voodoo magic at work. AMD has managed to improve performance in the last few domains where it was lagging behind Intel in the gaming and content creation space. All that remains is to see how Indian retailers will price the new Ryzen 9 5950X. The Intel Core i9-10900K is priced at ₹52,000 on Amazon and the AMD processor we've reviewed is priced at ₹72,000. You can see that there's little reason for getting the Intel desktop flagships anymore. There are still applications which make use of proprietary Intel tech where Team Blue continues to shine but for all other applications under the sun, AMD is the clear winner.

–Mithun Mohandas

SPECIFICATIONS
CORES: 12. | **THREADS:** 24 | **BASE CLOCK:** 3.8 GHz | **BOOST CLOCK:** 4.7 GHz | **PROCESS NODE:** 7nm | **L3 CACHE:** 64 MB | **L2 CACHE:** 512 KB per core | **TDP:** 105 W.

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